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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (MMD)(CBCS) Degree Examinations
January 2023

22MMD121 - CONTINUUM MECHANICS

Duration: 3 Hours

Max. Marks:100

Note: 1) Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

2) Missing data, if any may be suitably assumed.

Unit – I		Marks	BT*	CO*	PO*
1. a)	The strain components at a point in a body are $\epsilon_x=0.1$, $\epsilon_y=-0.05$, $\epsilon_z=0.05$, $\gamma_{xy}=0.3$, $\gamma_{yz}=0.1$, $\gamma_{xz}=-0.08$. Determine the strain invariants and the principal strains.	8	L*3	2	2
b)	Arrive at the cubic expression to determine the principal stress values and write the expression for the stress invariants.	12	L2	1	1
2. a)	Arrive at the equations of equilibrium in 3D assuming body forces are present.	12	L1	1	1
b)	Check whether the following expressions for the components of strain at a point are compatible or not. $\epsilon_x = 5 + x^2 + y^2 + x^4 + y^4$; $\epsilon_y = 6 + 3x^2 + 3y^2 + x^4 + y^4$; $\epsilon_z = \gamma_{xz} = \gamma_{yz} = 0$; $\gamma_{xy} = 10 + 4xy(x^2 + y^2 + 2)$	8	L2	2	2
3. a)	The known stress components at a point in a body, relative to the (x, y, z) axes are $\sigma_x = 20$, $\sigma_y = 10$; $\tau_{xy} = 30$, $\tau_{xz} = -10$ and $\tau_{yz} = 80$ MPa. Also, the second stress invariant is $I_2 = -7800$ (MPa) ² . Determine the stress component σ_z , the other stress invariants and the three principal stresses.	8	L2	1	2
b)	Arrive at the compatibility equations of strain in 3D.	12	L2	2	1
Unit – II					
4. a)	What do you mean by Airy's stress function? Check whether the following expressions are Airy's stress functions or not. (i) $\phi = Ax^2 + By^2$ (ii) $\phi = Ax^3$ (iii) $\phi = A(x^4 - 3x^2y^2)$	08	L2	3	2
b)	Arrive at the biharmonic equation for plane strain condition in the presence of body forces.	12	L2	3	1
5. a)	Derive Beltrami-Mitchel equation for plane stress condition assuming body forces are present.	8	L2	3	1
b)	Investigate what problem of plane stress is solved by the following stress function applied to the region included in $y=0$, $y=d$, $x=0$ on the side x-positive. $\phi = (3F/4d) \{xy - (xy^3/3d^2)\} + (P/2)y^2$	12	L3	3	2
6. a)	Using the inverse method, obtain the expressions for the stresses in a thin cantilever beam carrying a point load 'P' at its free end. The beam is of rectangular cross-section having width 'b' and depth '2h'	12	L2	3	1
b)	Explain in brief the application of Fourier series to solve two dimensional problems.	8	L1	3	1

P.T.O.

Unit – III

7. a) Arrive at the compatibility equations of strain in polar co-ordinate system.
- b) A thick cylinder of inner radius 10 cm and outer radius 15 cm is subjected to an internal pressure of 12 MPa. Determine the radial and hoop stresses in the cylinder at the inner and outer surfaces.
8. a) Fig. Q8a shows a two-cell tubular section as formed by a conventional airfoil shape, and having one interior web. An external torque of 10,000 Nm is acting in a clockwise direction. Determine the internal shear flow distribution. The cell areas are as follows:
 $A_1 = 680 \text{ cm}^2$, $A_2 = 2000 \text{ cm}^2$
 The peripheral lengths are indicated in Figure.

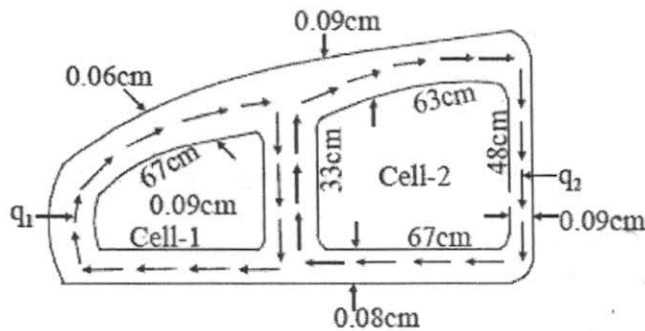


Fig. Q8a

- b) Explain the Kelvin and Maxwell's models used for modeling visco-elastic behaviour.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
